LEAP: Lead Exposure Assistance Program

Quantitative Funding Framework: Executive Summary

Municipal Infrastructure Finance Analysis

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1 The Problem

Columbus faces a federal mandate to replace all lead and galvanized private-side water service lines by 2037. The Lead Exposure Assistance Program (LEAP) offers homeowners a zero-interest, deferred-repayment loan of up to $10,000. Repayment occurs as a single lump sum when the property is sold, refinanced, or equity is extracted via HELOC. If inherited and never sold, repayment is deferred indefinitely — up to 99 years.

This creates an unusual financing challenge: the city must deploy tens of millions of dollars over twelve years with no predictable return schedule. We answer four questions:

1. How much upfront capital is required in 2026?
2. What funding level is sufficient with 95% probability?
3. How does uncertainty in participation, costs, and repayment affect the answer?
4. How does a loan structure compare to a grant?

2 Five-Phase Modeling Framework

We built five sequential models, each adding a layer of realism:

Phase	Type	Key Addition
1	Deterministic	Baseline cash flow model, sensitivity grid
2	Monte Carlo	Stochastic uncertainty, three breakage scenarios
3	Micro-simulation	Property heterogeneity, non-repayment quantification
4	Competing Risks	Explicit refi/HELOC channels, Vasicek interest rates
5	Meta-Distribution	Weighted 95% CI across all breakage scenarios

Data sources: Program pilot data (82 applicants, 42 completed loans, 63 contractor quotes) · WSL insurance records (2,862 policies) · System material cross-table (288,389 service line records) · CFPB HMDA 2024 (43,276 Columbus MSA mortgage records) · Census ACS B25026 (Columbus owner tenure) · SIAM-MTI simulated property dataset

3 Eligible Universe: 24,416 Properties

Customer-side pipe material (C) is unknown for 59,973 properties. We estimate hazardous properties ($\mathcal{H}$) among the unknowns using a **Bayesian conditional probability** approach. For each utility material category u, we compute the empirical hazard rate from properties where the customer-side material *is* known:

$$\hat{p}(u) = \frac{\sum_{j:\, U_j = u,\, C_j \in \mathcal{H}} n_j}{\sum_{j:\, U_j = u,\, C_j \neq \text{unknown}} n_j}$$

We then apply $\hat{p}(u)$ to each group of unknown-customer properties with the same utility material, treating the known rows as the prior evidence for what the unknowns are likely to be:

Utility Material	**Hazard Rate $\hat{p}(u)$**	**Est. Hazardous (unknowns)**
Lead / Galvanized	33%	4,701
Non-Lead Copper	10%	2,668
Unknown	17%	2,691
Other	<0.2%	3
Total		**10,062**

$$N_{\text{total}} = 14{,}354 \text{ (confirmed)} + 10{,}062 \text{ (estimated)} = \mathbf{24{,}416}$$

4 Phase 1: Deterministic Baseline

Using fixed assumptions — 51.22% participation (42/82 pilot), $7,582 average loan cost (+3%/yr), 12-year repayment lag — the deterministic model produces:

Scenario	**Required Capital**	**Repayments by 2037**
Base case ($L = 12$ yr)	$113,516,901	$0 (0%)
Optimistic ($L = 7$ yr)	$81,563,790	$33,878,654 (30%)

The critical insight: Under a 12-year average repayment lag, *not a single dollar returns before 2038*. The program behaves like a grant during its own deployment window — the loan structure provides long-run cost recovery but no near-term liquidity relief.

Sensitivity: Participation rate dominates all other variables. Moving from 33% to 89% uptake changes required capital by 2.7×. Beyond 12 years, additional repayment lag has *zero* impact on near-term capital needs.

Loan vs. Grant Comparison

Model	**Upfront Capital**	**Long-Run Net Cost**	**True Economic Cost**
Grant	$113,516,901	$113,516,901	$113,516,901
Loan ($L = 12$)	$113,516,901	$0	$47,677,098
Loan ($L = 7$)	$81,563,790	$0	$27,811,641

The loan program's true economic cost — the implicit financing subsidy foregone on operating reserves at 3.5%/yr — is \$47.7M, less than half the grant cost. Adding Phase 3's \$13M non-repayment loss gives a total true cost of approximately \$60.7M, or 53% of the grant equivalent.

5 Phase 2: Monte Carlo Simulation

Phase 2 replaces fixed assumptions with probability distributions and runs 10,000 simulations per scenario:

- **Universe size:** $\mathcal{N}(24{,}416,\ 1{,}006^2)$ — 10% uncertainty on the estimated component
- **Participation rate:** Beta(42, 40) — the Beta distribution naturally encodes the pilot outcome as 42 successes and 40 failures, giving a 95% CI of [40%, 62%]
- **Repayment lag:** LogNormal(mean=12, std=5 yr) — log-normal is appropriate for positive, right-skewed quantities
- **Replacement cost:** LogNormal fit to 63 pilot quotes. We verified goodness-of-fit using the Kolmogorov–Smirnov test, which compares the empirical and fitted CDFs; the resulting p-value of 0.41 indicates no statistical basis for rejecting the log-normal hypothesis

The Breakage Rate Problem

The insurance dataset (2,862 WSL policies, 41 repairs over 5 months) implies a 3.44% annual breakage rate — 14× higher than the pilot rate of 0.24%. The discrepancy has two causes: (1) insurance policyholders self-select because they suspect risk, and (2) the pilot denominator (all 24,416 eligible properties) is too large. We model three scenarios:

Scenario	Rate	Derivation	95th Pct Capital
Primary	0.24%	Pilot data; denominator too large	\$133,991,878
Adjusted	**1.38%**	Insurance ÷ 2.5× bias factor	**\$164,705,237**
Adverse	3.44%	Raw insurance rate; no correction	\$218,243,730

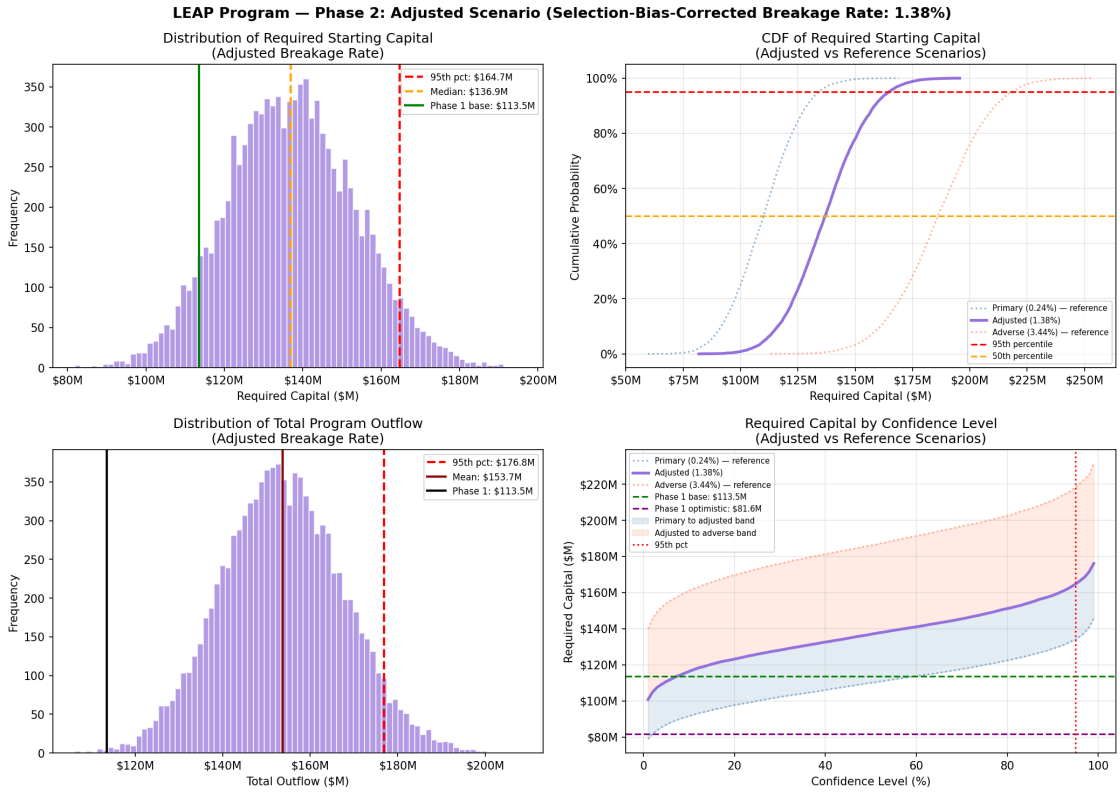

Figure 1: Phase 2 adjusted scenario (1.38% breakage, 10,000 simulations). The 95th percentile of $164.7M is our primary recommended funding level. Primary (0.24%) and adverse (3.44%) shown as dotted reference lines.

The Phase 1 estimate of $113.5M falls at the **40th percentile** of the primary distribution — insufficient 40.7% of the time. It is not a conservative number.

6 Phase 3: Property-Level Micro-Simulation

Note: Phase 3 uses a synthetically generated property dataset (SIAM-MTI). Results are structural illustrations.

Phase 3 models all 6,057 eligible properties individually. We use **logistic regression** to predict each property's participation probability. Logistic regression is appropriate here because the outcome is binary (participates or not) and we have multiple continuous and categorical predictors. The features and their expected signs are:

Feature	Direction	Rationale
Owner-occupied	↑	Direct benefit; dominant predictor
Pipe hazard score	↑	Greater urgency drives engagement
Assessed value	↓	High-value properties more likely to self-fund
Ownership length	↑	Long-tenure owners less likely to sell soon
Year built	↓	Newer properties less likely to have severe deterioration

All fitted coefficients have the expected sign. Phase 3's primary contribution is the quantification of **permanent non-repayment via inheritance**:

Property Type	Occupancy	Non-Repayment Rate	Mean Repayment Lag
Single family	Owner-occupied	10%	14 yr
Single family	Investor-owned	8%	10 yr
Duplex	Owner-occupied	15%	12 yr
Duplex	Investor-owned	10%	9 yr
Multi-family	Owner-occupied	20%	11 yr
Multi-family	Investor-owned	12%	8 yr
Blended		**11.57%**	

Mean permanent loss: $13.0M (12 cents per dollar loaned — never recovered)

Phase 3's 95th percentile capital of $102M appears lower than Phase 2 because it models only the voluntary participation channel. Its contribution is the $13M non-repayment finding, not a capital requirement comparison.

7 Phase 4: Competing Risks Repayment Model

Phases 1–3 modeled repayment as a single lag distribution, implicitly treating property sale as the only trigger. Phase 4 replaces this with a **competing risks framework** in which three triggers operate simultaneously and the loan repays when the *first* fires:

$$T_{\text{repay}} = \min(T_{\text{sale}},\ T_{\text{refi}},\ T_{\text{HELOC}})$$

Each trigger is calibrated to real Columbus data:

- **Property sale** — Weibull hazard (k=1.17, λ=13.9 yr) fit to Census ACS B25026. The Weibull is chosen because its hazard rate increases with tenure — homeowners who have lived in a property longer are more likely to sell in any given year. A 5-year LEAP selection dampening (40% of normal hazard) is applied, reflecting that recent borrowers signal long-term attachment.
- **Refinancing** — 2.33%/yr at 7.44% avg rate (CFPB HMDA 2024, 43,276 Columbus records). Rate sensitivity is modeled via an exponential function anchored to observed HMDA data.

- **HELOC** — 0.75%/yr at 7.44%, with interest rates drawn from a **Vasicek mean-reverting process** across three macro scenarios (low/base/high long-run equilibrium rates). A 2-year seasoning period applies before refi/HELOC channels activate.

The Headline Finding

Repayment Trigger	Share of All Repayments
Property sale	38.1%
Refinancing	**38.7%**
HELOC	8.6%
Never (non-repay)	14.6%

Refinancing accounts for nearly as many repayments as property sales — a channel entirely absent from Phases 1–3. **Fed rate policy directly affects LEAP cash flows.** A rate cut cycle reduces required capital by $30–44M relative to Phase 2.

Rate Scenario	95th Pct Capital	vs. Phase 2 Primary
Low rates	$89,779,476	−33%
Base rates	$97,067,155	−28%
High rates	$102,147,997	−24%
Rate sensitivity	**$12.4M**	

Phase 4 is lower than Phase 2 because the refi and HELOC channels bring repayments *into* the program window. Phase 2 remains the conservative planning figure if equity extraction activity is uncertain.

8 Phase 5: Weighted Meta-Distribution and Final Answer

We do not know which breakage scenario is correct. Phase 5 assigns probability weights to each Phase 2 scenario and combines 100,000 samples into a single mixture distribution — producing a true 95% CI that accounts for both within-scenario simulation uncertainty and between-scenario model uncertainty. Mixing distributions with spread means introduces fat tails, inflating the upper percentiles above what any single scenario alone would produce.

Scenario	Weight	Rationale
Primary (0.24%)	20%	Likely understated; denominator too large
Adjusted (1.38%)	**55%**	**Best estimate; real data with principled correction**
Adverse (3.44%)	25%	Receives weight because correction factor is itself assumed

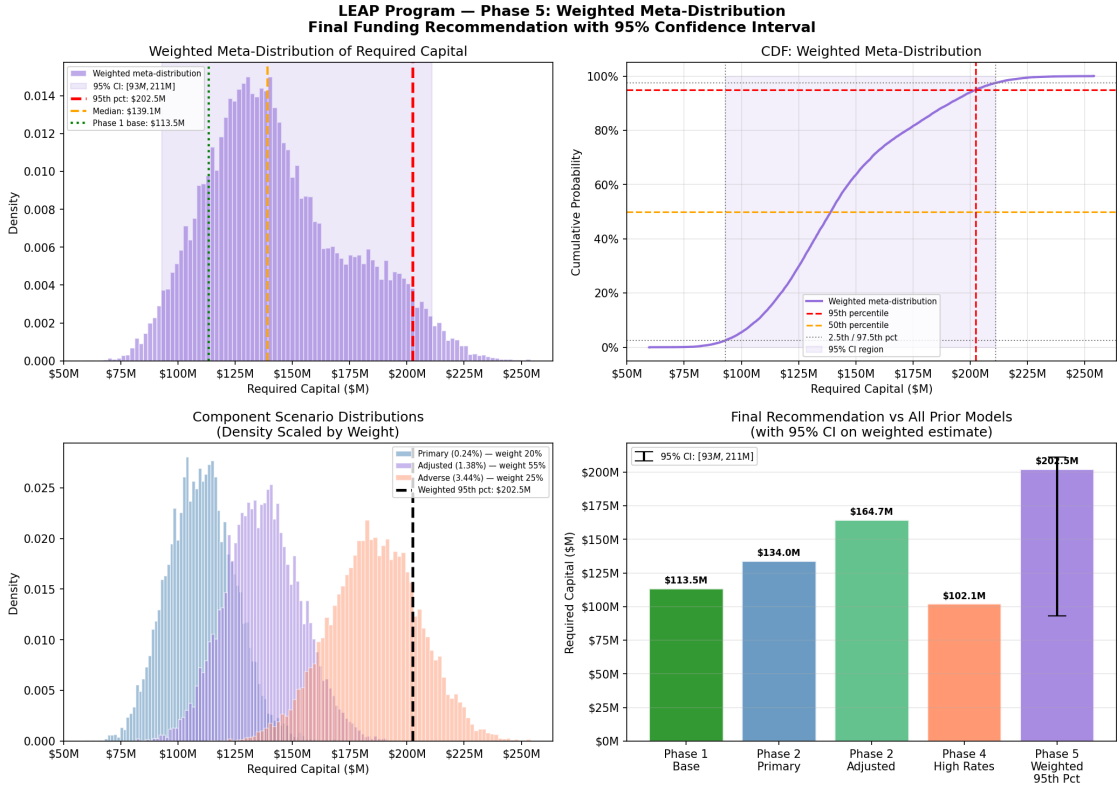

Figure 2: Phase 5 weighted meta-distribution (100,000 samples). The 95th percentile of \$202.5M and 95% CI [\$93M, \$211M] account for both simulation uncertainty and breakage rate model uncertainty. The \$37.5M gap between \$165M and \$202.5M is the quantified cost of not knowing the true pipe failure rate.

Metric	**Value**
Median	\$139,122,531
95th percentile	**\$202,459,838**
95% confidence interval	[\$93.0M, \$211.1M]

No defensible weighting scheme produces a number below \$165M. The \$37.5M gap between \$165M and \$202.5M is the cost of not knowing the true breakage rate — resolvable with more systematic data collection.

9 Final Recommendation

Tier	Amount	Meaning
Conditional floor	$102.1M	Phase 4: if refi/HELOC channels active
Absolute floor	$113.5M	Phase 1 deterministic; fails 40% of the time
Lower bound	$134.0M	Phase 2 primary; breakage likely understated
Recommended	**$164.7M**	**Phase 2 adjusted; 95% confidence, bias-corrected**
Risk-averse	$202.5M	Phase 5 weighted; 95% across all scenarios
Stress test	$218.2M	Phase 2 adverse; no selection bias correction

95% Confidence Interval (Phase 5): [$93M, $211M]

Policy Actions

1. **Fund $165M** as the baseline capital reserve. Consider $202.5M if the city wants 95% confidence even accounting for model uncertainty.
2. **Structure as loans, not grants.** True economic cost of loans ($60.7M) is 53% of the grant equivalent ($113.5M).
3. **Budget a $13M non-repayment reserve.** Permanent losses from inheritance are real and quantifiable.
4. **Prioritize participation outreach.** The dominant driver of capital needs — a 33% to 89% uptake shift changes required capital by 2.7×.
5. **Monitor housing turnover and Fed rate policy** annually as leading indicators of repayment timing.
6. **Collect more breakage data.** Worth up to $37.5M in reduced capital reserve requirements.